

Academic Writing for Master Students

How to Structure and Develop Your Master Thesis

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A Note on Context

- ▶ There's more than one correct way to do academic writing.
- ▶ The insights here are based on my own experience in conducting research in financial economics.
- ▶ ...and in supervising bachelor's and master's theses.
- ▶ I'll assume you're planning to write an **empirical** master thesis.
- ▶ Target audience: finance.
- ▶ (Minor) adjustments may apply:
 - ▶ Supervisor's preferences.
 - ▶ Literature review, theory, or math-focused theses.
- ▶ Interested in digging deeper? Check out the bibliography!

Table of Contents

Overview of Academic Writing

Structure of a Master Thesis

Writing Each Section

LaTeX Users Only

Best Practices

What Is the Goal of (Academic) Writing?

- ▶ Writing should allow people to comprehend *as* they read.
 - ▶ ×: Its tail wags the dog. The square of a number is not a prime number. That is what Jill learns when she opens her notebook. She opens her notebook because she likes to learn math when the dog is happy.
 - ▶ ✓: The dog wags its tail. And when it is happy, Jill likes to learn her math. So, she opens her notebook, and the first lesson she learns is that the square of a number is not a prime number.
- ▶ A formal style of writing used in universities and scholarly publications.
- ▶ Purpose: Communicate research findings and ideas clearly and effectively.
- ▶ Key traits: Clear structure, evidence-based arguments, and formal tone.

General Structure of a Master Thesis

- ▶ **Front Page**
- ▶ **Abstract**
- ▶ **Table of Contents**
- ▶ **Introduction**
- ▶ **Related Literature**
- ▶ **Data and Method**
- ▶ **Empirical Findings**
- ▶ **Conclusion**
- ▶ **References**
- ▶ **Appendix**

Front Page and Abstract

Front Page:

- ▶ Title, Author, Institution, Date.
- ▶ Statutory declaration.

Abstract:

- ▶ A concise and focused summary of your thesis.
- ▶ Include objectives, methods, findings, and significance.
- ▶ Many academic journals have 100-word limit, but less strict for MT (~ 12/15 lines)
- ▶ Last thing to write!

Table of Contents (1)

Table of Contents:

- ▶ Outline of thesis structure.
- ▶ Ensure logical order and accuracy.
- ▶ Numerical classification.

Things to avoid

- ▶ Avoid sub-sections in the Introduction
- ▶ Limit excessive subdivision (e.g., 3.4.1.1)
- ▶ Ensure subsections are justified by enough content
- ▶ Use concise, clear section headlines:
 - ▶ **NO!** 1.2.1. Changes in consumer behaviour occurring when there is an increase in the net median income of West Austrian [..]
 - ▶ **NO!** 2. Chapter Two
 - ▶ **NO!** 3.4 GDP in Austria has increased from 2003 to 2005

Table of Contents (2)

Sub-sections

YES!

- ▶ 1. Data and Summary Statistics
 - ▶ 1.1 Local Fama and French Factors
 - ▶ 1.2 Definition of Main Variables
 - ▶ 1.3 Summary Statistics
- ▶ 2. Life-Cycle Variation in the Value Tilt

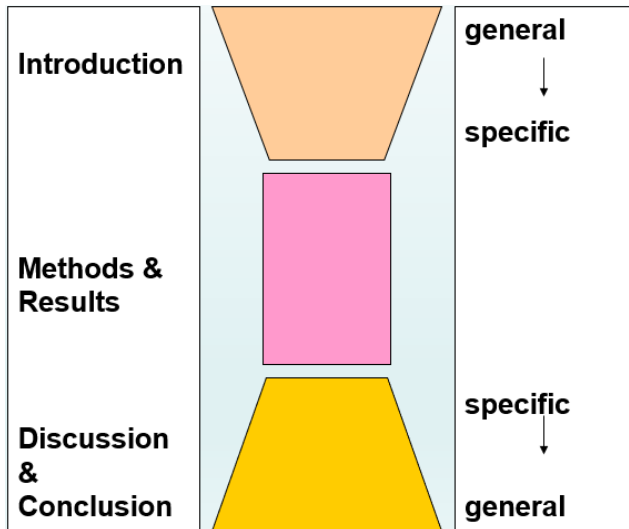
NO!

- ▶ 1. Data and Summary Statistics
 - ▶ 1.1 Local Fama and French Factors
- ▶ 2. Life-Cycle Variation in the Value Tilt

Table of Contents (3)

- ▶ **1. Introduction**
- ▶ **2. Literature Review**
 - ▶ 2.1 Bond Risk Premia
 - ▶ 2.2 Value, Carry and Momentum as Style Measures
- ▶ **3. Data and Methodology**
 - ▶ 3.1 Zero Coupon Yield Data
 - ▶ 3.2 Bond Excess Returns
 - ..
 - ▶ 3.5 Regression Specification
 - ▶ 3.6 Summary Statistics
- ▶ **4. Empirical Analysis**
 - ▶ 4.1 Principal Component Analysis
 - ..
- ▶ **5. Robustness Checks and Limitations**
- ▶ **6. Conclusion**

Hourglass Model by Swales



Introduction

- ▶ Very important!
- ▶ How to structure it effectively?
 - ▶ **Background:** *Why is this topic interesting?*
 - ▶ **Key research question:** What is the research gap?
 - ▶ **Solution/Approach:** How do you address it?
 - ▶ **Main results:** Briefly summarize your key findings (2–3 paragraphs).
 - ▶ **Contributions to the literature:** Highlight how your work adds value.
- ▶ Plan, write, and revise!
- ▶ Common pitfalls to avoid:
 - ▶ Research question is unclear.
 - ▶ Research question appears too late (e.g., on page 5).
- ▶ Keep it simple!
 - ▶ *Value added = output minus input.*

'The Term Structure of Currency Carry Trade Risk Premia', by Lustig, Andreas Stathopoulos, and Adrien Verdelhan, AER, 2019

Carry trades correspond to simple investment strategies that are funded by borrowing in low interest rate currencies and invest in high interest rate currencies. What are their expected returns? According to the uncovered interest rate parity (UIP) condition, which assumes that investors are risk-neutral, expected carry trade returns should be zero. Yet, empirically, borrowing in low interest rate currencies and investing in Treasury bills of high interest rate currencies delivers large excess returns. This is the UIP puzzle, and it gave rise to a large literature that studies the role of systematic risk and expectational errors in exchange rates. Our paper revisits the empirical evidence on carry trades and deepens the puzzle.

Our paper explores the properties of the same carry trade investment strategy implemented with long-term bonds, and compares it to the standard strategy that

'The Term Structure of Currency Carry Trade Risk Premia', by Lustig, Andreas Stathopoulos, and Adrien Verdelhan, AER, 2019

role of systematic risk and expectational errors in exchange rates. Our paper revisits the empirical evidence on carry trades and deepens the puzzle.

Our paper explores the properties of the same carry trade investment strategy implemented with long-term bonds, and compares it to the standard strategy that uses Treasury bills. We focus on the same set of G10 countries and consider the strongest predictors of bond and currency returns: the level and slope of the yield curve. The first strategy we consider goes long the bonds of high interest rate currencies and short the bonds of low interest rate currencies, whereas the second strategy goes long the bonds of flat yield curve currencies and short the bonds of steep yield curve currencies. Most importantly, the investment horizon is one month, as in the

'The Term Structure of Currency Carry Trade Risk Premia', by Lustig, Andreas Stathopoulos, and Adrien Verdelhan, AER, 2019

classic tests of the UIP condition, not ten years, as in the tests of the UIP condition over long horizons. We find that, as the maturity of the bonds increases, the average excess return declines to zero. In other words, whereas the carry trade implemented with Treasury bills is profitable, the carry trade implemented with long-term bonds is not. Similar results emerge in individual country time-series predictability tests: as the maturity of the bonds increases, the predictability of the cross-country differences in dollar bond returns disappears.

The downward-sloping term structure of carry trade risk premia that we uncover represents a challenge for the leading models in international finance. To illustrate

'The Term Structure of Currency Carry Trade Risk Premia', by Lustig, Andreas Stathopoulos, and Adrien Verdelhan, AER, 2019

premium on long-term bonds.

Our results are related to, but different from, the long-run UIP condition. The long-run UIP condition compares foreign and domestic long-term interest rates to long-term changes in exchange rates. Chinn and Meredith (2005) finds that long-run UIP is a potentially valid description of the data. However, empirical tests lack power in finite samples: intuitively, there are few non-overlapping observations of 10-year windows available so far. From no-arbitrage conditions, we show that long-run UIP always holds for temporary shocks and thus long-run UIP deviations have to come from permanent shocks to exchange rates. For long-run UIP to hold at

Note on the Language

- ▶ Never underestimate the importance of clear and concise writing! Convey your message as clearly and simply as possible.
- ▶ Keep your sentences short and avoid repetition.
- ▶ Your writing style should generally be objective. Avoid personal views unless necessary, and ensure they are clearly identified as such.
- ▶ Use **simple present tense** when reporting:
 - ▶ *“Prior work, from Campbell and Shiller (1991) to Bekaert and Hodrick (2001) and Bekaert, Wei, and Xing (2007), **shows** that investors earn higher returns.”*
 - ▶ *“Finally, we **collect** data on inflation rates and sovereign credit ratings [...].”*
- ▶ Writing is like math – you learn it by doing it.

Related Literature

- ▶ Answer the question: *“What do we already know about this topic?”*
- ▶ Review existing studies.
- ▶ Highlight the gap your thesis will address.
- ▶ Start with seminal papers in the field.
- ▶ Tip: Keep your literature review concise (around 4 pages).

How to Work with Literature?

- ▶ Use databases like Google Scholar, and SSRN to find relevant papers.
- ▶ Focus on peer-reviewed journals to ensure quality and reliability.
- ▶ Organize your literature review thematically or methodologically.
- ▶ Use reference management tools (e.g., EndNote, BibTex) for efficient citation.

Export citation (example)



ARTICLE | Open Access |

Pricing Currency Risks

MIKHAIL CHERNOV, MAGNUS DAHLQUIST LARS LOCHSTOER

First published: 16 November 2022 | <https://doi.org/10.1111/jofi.13190> | Citations: 1

SECTIONS

PDF

TOOLS

SHARE

ABSTRACT

The currency market features a small cross-section of assets that can be characterized by few signals: interest differentials and exchange rates. We exploit these properties to construct the ex ante optimal portfolio of individual currencies. The portfolio is updated in real time using currency trading strategies, conditionally and unconditionally. Our results show that assets that does not affect their risk premiums in the short run. Expansions of carry strategies based on intermediary capital or global volatility are related to these unpriced components, while consumption growth is related to the priced component of returns.

IN THIS PAPER, WE ARGUE that research on cross-sectional (CS) currency pricing can depart fruitfully from the faster mining path established by the equity literature. That is, because a



Figures Refe

Recommend

[Basics of Curre](#)

Shani Shamah

[Encyclopedia of](#)

[PRICING CURRE
STOCHASTIC IN
DIFFUSION PRO](#)

Ako Doffou, Jimm

[Journal of Finan](#)

[Empirical perfo
pricing models.](#)

How to Assess the Quality of a Paper?

- ▶ **Check the publication venue:**

- ▶ Top finance journals include: *Journal of Finance (JF)*, *Review of Financial Studies (RFS)*, *Journal of Financial Economics (JFE)*, *Review of Finance (RoF)*, *Journal of Financial and Quantitative Analysis (JFQA)*.

- ▶ **Working papers:** High quality if presented at top conferences (e.g., AFA, WFA, EFA, Cavalcade, FIRS) or authored by respected scholars.

- ▶ **Use journal rankings:** Refer to resources like the ABS Journal Ranking list for guidance.

- ▶ **When in doubt:** Always consult your supervisor.

- ▶ **Avoid confirmation bias:** Don't settle for papers that merely confirm what you were looking for without further scrutiny.

Data and Methods

Data and Methods

- ▶ Clearly explain: *“How did you get the data and reach your findings?”*
- ▶ Describe datasets, sources, and analytical tools.
- ▶ Include summary statistics for key variables to help readers understand your data and evaluate your approach.
- ▶ Be precise—your methods should be replicable. Think of it like a recipe: list all “ingredients,” their sources, and clear “instructions.”
- ▶ Maintain consistent notation and variable labels (Return_{10Y}^t , rx_t^{10Y} , $\text{Ret10Yr}(t)$).
- ▶ Be precise on the notation (e.g., rx_{t+1}^{10Y} , $rx_{i,t+1}^{10Y}$, rx_t^{10Y} , etc.).
- ▶ Start early! Data collection, cleaning, and preparation can take longer than expected.

Empirical Findings

Empirical Findings:

Your results are the centerpiece of your thesis. Present them clearly and effectively to answer the question: *What are your findings?*

Tips for Presentation:

- ▶ Ensure axis labels are readable and avoid overlaps in figures.
- ▶ Use clear and interpretable scales for all axes.
- ▶ Add captions to tables and figures to describe variables/data shown.

TABLE 1—TIME-SERIES PREDICTABILITY REGRESSIONS (*Continued*)

	Bond dollar return diff.		Currency excess return		Bond local currency return diff.		Slope diff.	Observations
	$rx^{(10),\$}$	$-rx^{(10)}$	rx^{FX}		$rx^{(10),*}$	$-rx^{(10)}$		
	β	$R^2(\%)$	β	$R^2(\%)$	β	$R^2(\%)$	p -value	
<i>Panel B. Yield curve slopes</i>								
Australia	3.84 (1.69)	1.54	-1.00 (1.16)	-0.02	4.84 (0.96)	7.65	0.03	492
Canada	4.04 (1.23)	2.25	-0.72 (0.79)	-0.07	4.76 (0.81)	9.09	0.00	492
Germany	0.50 (1.57)	-0.18	-3.05 (1.37)	1.15	3.55 (0.82)	4.07	0.11	492
Japan	-0.32 (1.12)	-0.19	-4.18 (0.94)	2.91	3.85 (0.81)	3.96	0.02	492
New Zealand	2.94 (2.35)	1.26	-1.60 (1.28)	0.62	4.55 (1.41)	7.41	0.11	492
Norway	0.59 (0.98)	-0.12	-2.03 (0.97)	1.33	2.62 (0.52)	3.35	0.07	492
Sweden	3.12 (1.21)	2.12	-0.13 (1.02)	-0.20	3.25 (0.82)	5.29	0.06	492
Switzerland	0.97 (1.05)	-0.06	-3.59 (1.27)	1.97	4.55 (1.00)	8.82	0.01	492
United Kingdom	1.59 (1.28)	0.17	-3.17 (1.62)	2.11	4.75 (0.85)	7.95	0.03	492
Panel	1.94 (0.96)	0.42	-2.02 (0.82)	0.83	3.96 (0.66)	6.08	0.00	4,428
Joint zero p -value	0.08		0.01		0.00		0.00	

Notes: The table reports regression results of the bond dollar return difference ($rx_{t+1}^{(10),\$} - rx_{t+1}^{(10)}$, left panel) or the currency excess return (rx_{t+1}^{FX} , middle panel) or the bond local currency return difference ($rx_{t+1}^{(10),*} - rx_{t+1}^{(10)}$, right panel) on the difference between the foreign nominal interest rate and the US nominal interest rate ($r_{t+1}^f - r_t^f$, panel A) or the difference between the foreign nominal yield curve slope and the US nominal yield curve slope ($[y_t^{(10),*} - y_t^{(1,*)}] - [y_t^{(10)} - y_t^{(1)}]$, panel B). The column *Slope diff.* presents the p -value of the test for equality between the slope coefficient in the bond dollar return difference regression and the slope coefficient in the currency excess return regression for each country. The last line in each panel presents the p -value of the joint test that all individual-country regression coefficients in the respective column are zero. We use returns on 10-year coupon bonds. The holding period is one month and returns are sampled monthly. The log returns and the yield curve slope differentials are annualized. The sample period is 1975:1–2015:12. The balanced panel consists of Australia, Canada, Japan, Germany, Norway, New Zealand, Sweden, Switzerland, and the United Kingdom. In individual country regressions, standard errors (in parentheses) are obtained with a Newey and West (1987) approximation of the spectral density matrix, with the lag truncation parameter (kernel bandwidth) equal to 29. Panel regressions include country fixed effects, and standard errors (in parentheses) are obtained using the Driscoll and Kraay (1998) methodology, with the lag truncation parameter (kernel bandwidth) equal to 29. All p -values are fixed- b p -values, calculated using the approximation of the corresponding fixed- b asymptotic distribution in Vogelsang (2011, 2012).

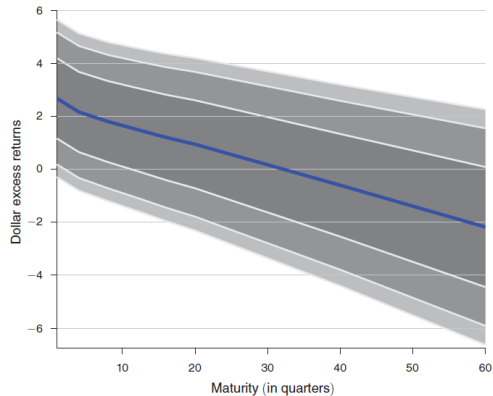


FIGURE 2. LONG-MINUS-SHORT FOREIGN BOND RISK PREMIA IN US DOLLARS

Notes: The figure shows the dollar log excess returns as a function of the bond maturities. Dollar excess returns correspond to the holding period returns expressed in US dollars of investment strategies that go long and short foreign bonds of different countries. The unbalanced panel of countries consists of Australia, Canada, Japan, Germany, Norway, New Zealand, Sweden, Switzerland, and the United Kingdom. At each date t , the countries are sorted by the slope of their yield curves into three portfolios. The first portfolio contains countries with flat yield curves while the last portfolio contains countries with steep yield curves. The slope of the yield curve is measured by the difference between the ten-year yield and the three-month interest rate at date t . The holding period is one quarter. The returns are annualized. The dark shaded area corresponds to one-standard-error bands around the point estimates. The gray and light gray shaded areas correspond to the 90 percent and 95 percent confidence intervals. Standard deviations are obtained by bootstrapping 10,000 samples of non-overlapping returns. Zero-coupon data are monthly, and the sample window is 1985:4–2015:12.

Conclusion

- ▶ Follow the Hourglass Model: specific $\rightarrow$ general.
- ▶ Summarize briefly (!) findings and contributions.
- ▶ Message you want the reader to take away from reading the paper.
- ▶ Suggest areas for future research.

Structuring a Thesis in LaTeX

Organize your content:

- ▶ Use a master file to include the main body:
 - ▶ `main.tex`
- ▶ Keep tables and figures in separate files:
 - ▶ `\input{tables_latest}`
 - ▶ `\input{figures_latest}`

Using Bibliography in LaTeX

- ▶ Use BibTeX to manage references:
 - ▶ Include the natbib package: `\usepackage{natbib}`
 - ▶ Set the bibliography style: `\bibliographystyle{chicago}`
 - ▶ Add your bibliography file: `\bibliography{your_bib_file}`
- ▶ Ensure all cited works are relevant and up-to-date.
- ▶ Classify entries using BibTeX types, e.g.:
 - ▶ `@article{}`
 - ▶ `@unpublished{}`
 - ▶ `@techreport{}`

Writing Tips

- ▶ Start early and plan your work carefully.
- ▶ Rough guide: 40/50 pages including bibliography but excluding Appendix.
- ▶ Use clear, concise, and formal language.
- ▶ Cite all sources properly to avoid plagiarism.
- ▶ Keep paragraphs focused on one main point and avoid overly long ones.
- ▶ Get feedback: ask a peer to look at your thesis and check, for instance:
 - ▶ that they see the order of the sections and subsections logical
 - ▶ they walk away with the right main message.

A Note on Planning

- ▶ Develop a daily plan or experiment to find a schedule that works best for you.
- ▶ Try techniques like the [Pomodoro Method](#): focus for 25 minutes, take a 5-minute break, and after 4 sessions (2 hours), take a longer break. Use [online timers](#) or adjust timings as needed.
- ▶ Share tips with others but avoid comparisons—what works varies for everyone.

Bibliography

References

- ▶ Chaubey, Varanya (2018), *The Little Book of Research Writing: The Structural Challenge of Communicating Knowledge + a Method to Meet it*, CreateSpace Independent Publishing Platform.
- ▶ Creme, Phyllis; Lea, Mary R. (2003), *Writing at University: A Guide for Students*, 2nd ed., Maidenhead: Open University Press.
- ▶ Flowerdew, John (ed.) (2002a), *Academic Discourse*, Harlow etc.: Longman.
- ▶ Herles, Martin, *Slides on Academic Writing*.
- ▶ Mautner, Gerlinde; Ross, Christopher (2023), *English Academic Writing – A Guide for the Humanities and Social Sciences*, Munich: UVK.
- ▶ Wallwork, Adrian (2011), *English for Writing Research Papers*, New York: Springer.